

Defence Institute of Advanced Technology, Pune

Course Name: AM 607, Assessment-2, Answer All questions.

1. Explain the logistic regression with mathematical expressions. If the decision boundary learnt by the logistic regression takes the form $f(x,y,z) = 3x + 2y - 7z + 10$, then name the following samples which belong to the same category. $A=[1,1,3]$, $B=[1,1,1]$, $C=[5,0,1]$, $D=[0,0,1]$
2. Explain or Justify the scenario where Accuracy is unreliable as a performance metric and what is the alternative metric that addresses the shortcomings of Accuracy.
3. Explain what is the impact of noise or outliers in learning decision trees, and how it can be addressed
4. True or False: (a). KNN cannot be used for regression. (b). Logistic regression is used for regression (c). Which classification algorithm works better when features are categorical? (d). The performance of KNN does not depend on the similarity measure. (e). Decision Trees cannot be built for multi-label classification